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CHEMISTRY - 50

# IR SPECTROSCOPY

* Write down the eqⁿ for the vibrational energy of a simple harmonic oscillator (a harmonic diatomic molecule).

The eqⁿ for the vibrational energy of a simple harmonic oscillator is given by,

$$\mathcal{E}_v = (v + \tfrac{1}{2})\,\bar{\omega}_{osc} \quad cm^{-1}$$

where, $v$ is the vibrational quantum number & $\bar{\omega}_{osc}$ is the oscillation frequency in $cm^{-1}$.

* Show that, a diatomic molecule can never have zero vibrational energy.

The oscillation frequency of an elastic bond of a diatomic molecule is given by,

$$\omega_{osc} = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}} \quad Hz \quad \text{—— ①}$$

where, $\mu$ is reduced mass & $k$ is force constant.
Dividing eqⁿ ① by the speed of light, $c$ to convert $\omega_{osc}$ in wave number,

$$\bar{\omega}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \quad cm^{-1}$$

Now, from Schrodinger eqⁿ, the allowed vibrational energies for any particular system is given by,

$$E_v = (v + \tfrac{1}{2})\,h\,\omega_{osc} \quad Joules \quad [v = 0, 1, 2 \ldots]$$

—— ②

Here, $v$ = vibrational quantum number.

Converting eqn (ii) into spectroscopic units,

$$\epsilon_v = \frac{E_v}{hc}$$

$$= \frac{(v + \frac{1}{2})\, h \cdot \omega_{osc}}{hc} \quad \text{Joules}$$

$$\epsilon_v = (v + \tfrac{1}{2})\, \bar{\omega}_{osc} \quad cm^{-1} \quad \text{——— (iii)}$$

* Eqn (iii) is known to be the eqn for the vibrational energy of a simple harmonic oscillator.

At the lowest vibrational energy, i.e, when $v = 0$,

eqn (ii) $\Rightarrow E_0 = \frac{1}{2} h\, \omega_{osc}$ Joules

eqn (iii) $\Rightarrow \epsilon_0 = \frac{1}{2} \bar{\omega}_{osc}$ $cm^{-1}$

That is, the diatomic or any molecule can never have zero vibrational energy indicating that, the atoms can never be completely at rest relative to each other.

* What is the selection rule for harmonic oscillator undergoing vibrational changes?

The selection rule: $\Delta v = \pm 1$

* Write & compare the energy of anharmonic oscillator & harmonic oscillator.

The energy of harmonic oscillator:

$$\epsilon_v = (v + \tfrac{1}{2})\, \bar{\omega}_{osc} \quad cm^{-1} \quad \text{——— (i)}$$

For anharmonic oscillator:

$$\epsilon_v = \bar{\omega}_e \left[1 - (v + \tfrac{1}{2})\, x_e\right](v + \tfrac{1}{2}) \quad cm^{-1} \qquad \text{—— (ii)}$$

Comparing eq$^n$ (i) & (ii),

$$\bar{\omega}_{osc} = \bar{\omega}_e \left[1 - (v + \tfrac{1}{2})\, x_e\right] \qquad \text{—— (iii)}$$

That is, the anharmonic oscillator behaves like harmonic oscillator with an oscillation ~~en~~ frequency that decreases with increasing $v$.

* Sketch schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic & anharmonic oscillators. Show the allowed transitions & expected spectrum.

A schematic diagram to represent the energy levels of a diatomic molecule regarding as a harmonic oscillator is —

For harmonic oscillators: $\epsilon_v = (v + \frac{1}{2})\bar{\omega}_{osc}$ $cm^{-1}$. And the selection rule is:

$$\Delta v = \pm 1$$

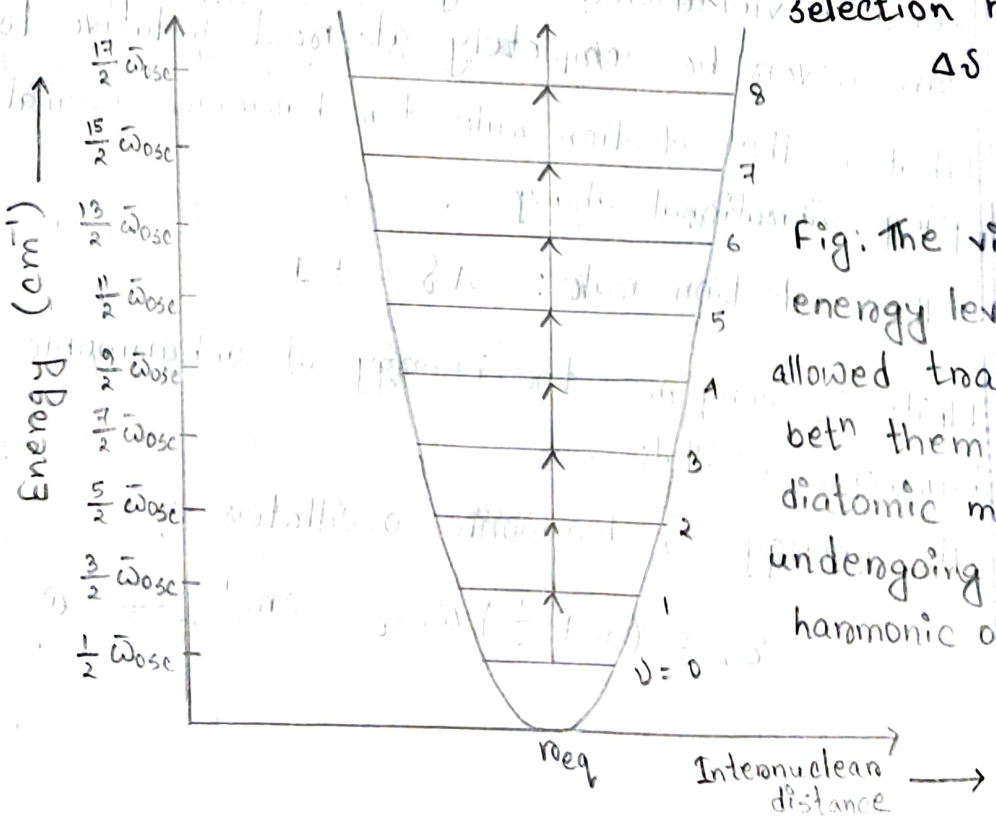
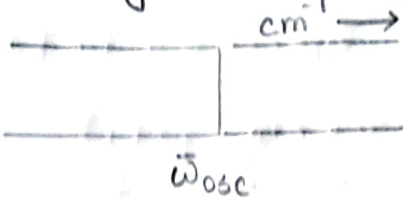



Fig: The vibrational energy levels & allowed transitions betn them for a diatomic molecule undergoing simple harmonic oscillation

The spectrum arising from the transition is-




For anharmonic oscillator :

$$\epsilon_v = \bar{\omega}_e \{1 - x_e(v + \tfrac{1}{2})\}(v + \tfrac{1}{2}) \quad \&$$

selection rule : $\Delta v = \pm 1, \pm 2, \pm 3, \ldots$

A schematic diagram for anharmonic oscillator is-

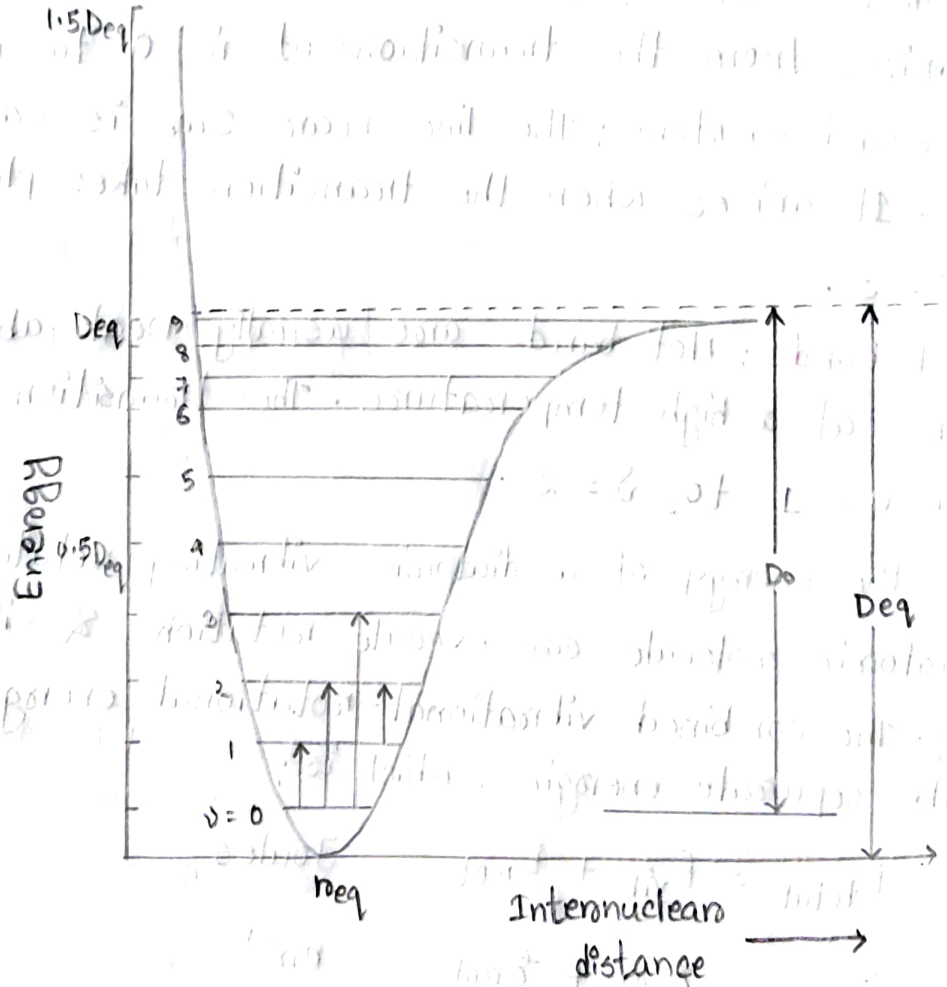
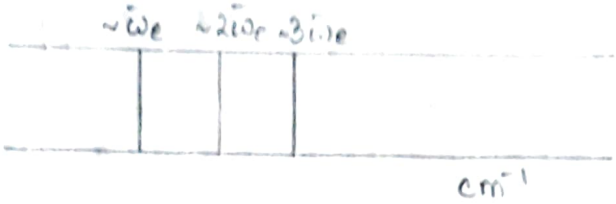



Fig: The vibrational energy levels & some transitions betⁿ the for a diatomic molecule undergoing anharmonic oscillation

The spectrum arising from the transitions —




The bands observed by these transitions—

1. Fundamental absorption: Fundamental absorption arises when the transition occurs from $\upsilon = 0$ to $\upsilon = 1$. The line near $\bar{\omega}_e$ is called fundamental absorption.

2. First overtone: The line near $2\bar{\omega}_e$ is called first overtone. It arises from the transition of $\upsilon = 0$ to $\upsilon = 2$.

3. Second overtone: The line near $3\bar{\omega}_e$ is called 2nd overtone. It arises when the transition takes place from $\upsilon = 0$ to $\upsilon = 3$.

4. Hot bands: Hot bands are usually weak absorptions which occurs at a high temperature. The transition takes place from $\upsilon = 1$ to $\upsilon = 2$.

* Write the energy of a diatomic vibrating rotator. Book-63 pg

A diatomic molecule can execute rotations & vibrations independently. The combined vibrational-rotational energy is the sum of the separate energies, which is:

$$E_{total} = E_{vib} + E_{rot} \quad \text{Joules}$$

$$\mathcal{E}_{total} = \mathcal{E}_{vib} + \mathcal{E}_{rot} \quad cm^{-1}$$

Taking separate expressions for $\mathcal{E}_{vib}$ & $\mathcal{E}_{rot}$, we have:

Now, $\varepsilon_v = (v+\frac{1}{2})\bar{\omega}_e - x_e(v+\frac{1}{2})^2\bar{\omega}_e$ $cm^{-1}$ ——— (i)

& $\varepsilon_J = BJ(J+1) - DJ^2(J+1)^2 + HJ^3(J+1)^3 + \dots$ ——— (ii)

By ignoring the ~~con~~small centrifugal distortion constants D, H etc. from eq$^n$ (ii), we can write the energy expression as –

$$\varepsilon_{J,v} = \varepsilon_J + \varepsilon_v$$

$$\varepsilon_{J,v} = BJ(J+1) + (v+\tfrac{1}{2})\bar{\omega}_e - x_e(v+\tfrac{1}{2})^2\bar{\omega}_e = \varepsilon_{total} \quad \text{——— (iii)}$$

Eq$^n$ (iii) is the energy expression for a diatomic vibrating rotator.

* How do you observe the O, P, Q, R, S bands for a diatomic vibrating rotator ? Or, explain the origin of O, P, Q, R, S.

The energy of a diatomic vibrating rotator is –

$$\varepsilon_{total} = BJ(J+1) + (v+\tfrac{1}{2})\bar{\omega}_e - (v+\tfrac{1}{2})^2\bar{\omega}_e x_e$$

& the selection rule : $\Delta v = \pm 1, \pm 2, \dots$ & $\Delta J = \pm 1$

Considering the $v=0 \longrightarrow v=1$ transition, we have,

$$\Delta\varepsilon_{J,v} = \varepsilon_{J',v=1} - \varepsilon_{J'',v=0}$$

$$= \left[BJ'(J'+1) + \frac{3}{2}\bar{\omega}_e - \frac{9}{4}\bar{\omega}_e x_e\right] - \left[BJ''(J''+1) + \frac{1}{2}\bar{\omega}_e - \frac{1}{4}\bar{\omega}_e x_e\right]$$

$$= BJ'(J'+1) - BJ''(J''+1) + \bar{\omega}_e - 2\bar{\omega}_e x_e$$

$$= B(J'^2 + J') - B(J''^2 + J'') + \bar{\omega}_e - 2\bar{\omega}_e x_e$$

$$= \bar{\omega}_e(1-2x_e) + B(J'^2 + J' - J''^2 - J'')$$

∴ Let, $\bar{\omega}_e(1-2x_e) = \bar{\omega}_0$ ,

$$\Delta \xi_{J,v} = \bar{\omega}_0 + B\left[(J' - J'') + (J'^2 - J''^2)\right]$$

$$= \bar{\omega}_0 + B\left[(J' - J'') + (J' - J'')(J' + J'')\right]$$

$$= \bar{\omega}_0 + B(J' - J'')(J' + J'' + 1) \quad \text{——①}$$

Here, $\bar{\omega}_0$ is the band origin.

Now, we can have 2 cases.

Case-1 : $\Delta J = +1$

$$\Rightarrow J' - J'' = +1 \quad \Rightarrow J' = J'' + 1$$

$$\therefore ① \Rightarrow \Delta \xi_{J,v} = \bar{\omega}_0 + B \cdot 1 \,(J'' + 1 + J'' + 1)$$

$$= \bar{\omega}_0 + 2B(J'' + 1) \quad \text{——②}$$

Case-2 : $\Delta J = -1$

$$\Rightarrow J' - J'' = -1 \quad \Rightarrow J'' = J' + 1$$

$$\therefore ① \Rightarrow \Delta \xi_{J,v} = \bar{\omega}_0 + B \cdot (-1) \cdot (J' + J' + 1 + 1)$$

$$= \bar{\omega}_0 - 2B(J' + 1) \quad \text{——③}$$

Combining these two eq$^n$s,

$$\Delta \xi_{J,v} = \bar{\omega}_0 + 2Bm \ \text{cm}^{-1} \quad ; \text{where } m = \pm 1, \pm 2 \ldots$$

——④

Eq$^n$ ④ represents the combined vibration-rotation spectrum, having equally spaced lines on each side of origin $\bar{\omega}_0$. Lines to the low frequency side of $\bar{\omega}_0$, corresponding to neg m ($\Delta J = -1$) are called the P branch. Lines to the high frequency side of $\bar{\omega}_0$, where m is +v &

$\Delta J = +1$ are called R branch. Lines arising for the transition $\Delta J = 0$ are called Q-branch. Similarly, when $\Delta J = -2$, the lines are called O-branch and when $\Delta J = +2$, the lines are called S-branch. That is,

| $\Delta J = -2$ | $-1$ | $0$ | $+1$ | $+2$ | |
|---|---|---|---|---|---|
| O | P | Q | R | S | branch |

That is how the branches are originated in a diatomic vibrating rotator.

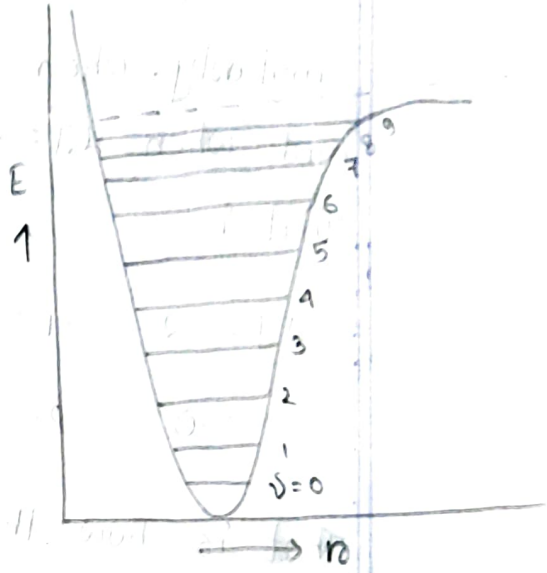
* P-branch – P-branch refers to a specific set of spectral lines observed in the diatomic vibrating rotator. These lines arises when $\Delta J = -1$, that is, lines to the low frequency side of band origin are called P-branch.

Q-branch : It also refers to a set of spectral lines that appear in the vibrational-rotational spectra when there is no change in the rotational quantum number J, i.e., $\Delta J = 0$.

R-branch : This is a set of spectral lines that are observed when $\Delta J = +1$, i.e., lines to the high frequency side of band origin are called R-branch.

* Explain why does spacing betⁿ vibrational levels in a Morse curve decreases gradually.

In a Morse curve, the spacing betⁿ vibrational levels decrease gradually.

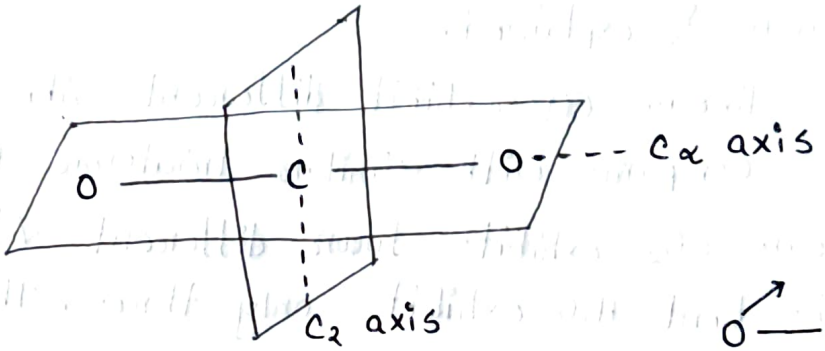


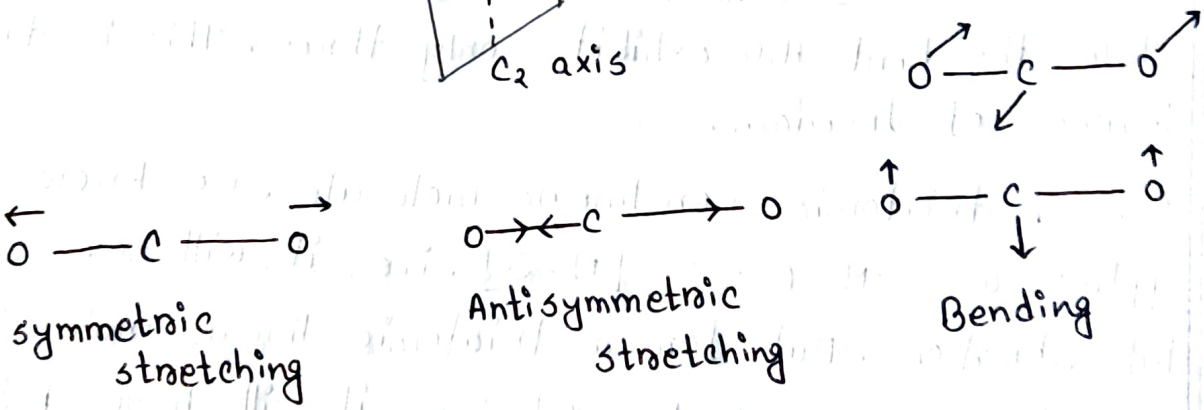

The Morse curve here represents how molecules vibrate. At lower energies, the vibrational levels are evenly apart, as the molecule behaves as a perfect spring. But as the molecule vibrates more & more, the bonds betⁿ the atoms stretch & does not behave like a perfect spring anymore. The further it stretches, the less energy it is needed to make it vibrate. Hence the energy differences betⁿ the vibrational levels get smaller as we go up to higher levels, meaning that the levels are packed more closely together.

Therefore, as the vibrations get stronger, the bond length increases due to stretching & so the energy levels get closer. Hence, the spacing betⁿ vibrational levels in a Morse curve decreases gradually.

* Sketch the symmetry & fundamental vibrations of $CO_2$ & explain them.

$CO_2$ is a linear triatomic molecule. For this molecule, there are 2 different sets of symmetry axes. There is an infinite numbers of $C_2$ axes passing through the C atom at right angles to the bond direction & an $C_\infty$ axis passing through the bond axis itself. The symmetry & fundamental vibrations of $CO_2$ are drawn below –




Since $CO_2$ is a linear triatomic molecule, it has $3N-5 = 4$ $[N=3]$ fundamental vibration modes; which are –

1. Symmetric Stretch ($\nu_1$): Both O atoms move simultaneously toward or away from C atom along bond axis. There is no dipole change, hence IR inactive.

2. Assymetric Stretch ($\upsilon_3$): One O moves toward C while other moves away. It is IR active as the dipole moment changes.

3. Bending ($\upsilon_2$): There are 2 degenerate bending modes — one in the plane of the paper & the other is in which the O atoms move simultaneously in & out of the plane. It is IR active.

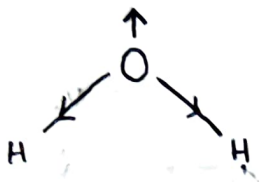
Therefore, the symmetry & fundamental vibrations of $CO_2$ are drawn & explained.

* Why does linear $CO_2$ exhibit different vibration modes than expected? Compare with similar triatomic $H_2O$.

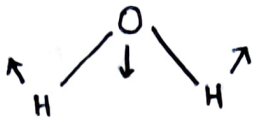
The linear $CO_2$ exhibits four different vibration modes while the bent $H_2O$ exhibits only three. This is due to their degrees of freedom.

For a triatomic non-linear molecule, we know the rule applies as $3N-6 = 3$ $[N=3]$, i.e, it will have 3 fundamental vibrations. But for a triatomic linear molecule, the rule is $3N-5 = 4$ $[N=3]$, i.e, it will have 4 fundamental vibrations. This is because in linear triatomic molecule, the bending mode alone consists of two vibrations- one in the plane of the paper & another in which 2 of the 3 atoms move simultaneously in & out of the plane. These 2 different vibrations in the bending mode are termed as degenerate. And because of this degeneracy, a linear molecule like $CO_2$ has an extra mode of vibration than a non-

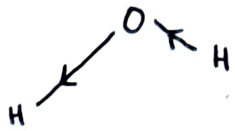
linear molecule like $H_2O$. The different vibrations of $H_2O$ & $CO_2$ are given below –



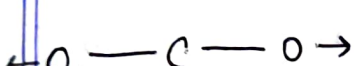
Symmetric stretch



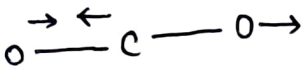
Symmetric Bending



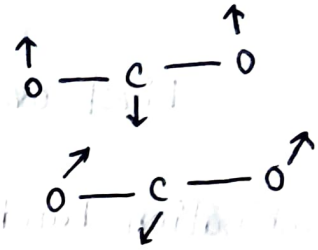
Asymmetric stretch



Symmetric stretch



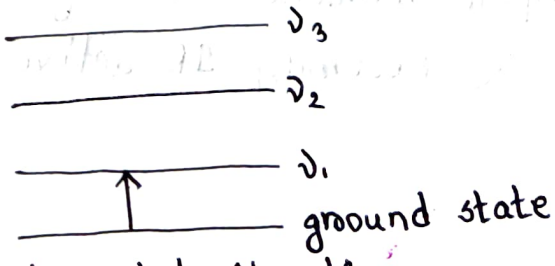
Asymmetric stretch



Bending

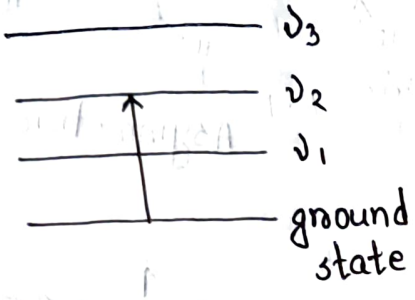
Therefore, the degeneracy of bending mode in $CO_2$ caus an extra vibration. And because of the absence of this degeneracy, the bent $H_2O$ cannot have the fourth vibration other than symmetric, asymmetric & bending stretching.

* Fundamental vibration corresponds to the lowest energ vibrational transition where the molecule moves from its ground state to the first excited vibrational state

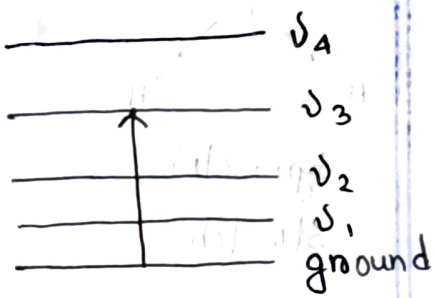



Fundamental vibration

* Overtones refer to a spectral band that occurs at a frequency which is a multiple of a fundamental vibrational frequency.



First overtone



$2^{nd}$ overtone

* Combination bands : When two fundamental vibrational frequencies in a molecule couple $(\nu_1 + \nu_2)$ to give rise to a new vibrational frequency within the molecule, it is known as combination band.

* State the basic criteria for a substance to be IR active with example.

For a substance to be IR active, it must have a change in the dipole moment during molecular vibrations. IR radiation is absorbed by the molecule when the vibration causes a change in dipole moment. For example : water molecule. The asymmetric stretching & bending modes of $H_2O$ cause change in dipole moment, allowing the molecule to absorb IR radiation & becoming IR active.

* Calculate the force constant of $H^{35}Cl$ molecule when the fundamental frequency of vibration for $H^{35}Cl$ is $2886\ cm^{-1}$. Molar mass of $^{35}Cl = 34.97$ amu.

We know,

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \quad \text{——①}$$

where, $\bar{\nu}_{osc}$ = wave number, $c$ = speed of light,

$\mu$ = reduced mass, $k$ = force constant

Given,

$\bar{\nu}_{osc} = 2886\ cm^{-1}$

Now, $\mu = \frac{m_H \cdot m_{^{35}Cl}}{m_H + m_{^{35}Cl}}$

$= \frac{1.008 \cdot 34.97}{1.008 + 34.97}$ amu

$= 0.9798$ amu

Putting the values in eqn ①,

$$2886 = \frac{1}{2\pi \times 3 \times 10^{10}} \times \sqrt{\frac{k}{0.9798 \times 1.67 \times 10^{-27}}}$$

$\Rightarrow k = 484.23\ kg/s^2$ (Ans)

* The fundamental band of CO in inert $CCl_4$ solvent is found at $2143\ cm^{-1}$. Calculate force constant.

We know,

$$\bar{\nu}_{osc} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}} \quad ; \text{ Given, } \bar{\nu}_{osc} = 2143\ cm^{-1}$$

Now,

$$\text{Reduced mass}, \mu = \frac{m_C \cdot m_O}{m_C + m_O} \times 1.67 \times 10^{-27} \text{ kg}$$

$$= \frac{12 \cdot 16}{12 + 16} \times 1.67 \times 10^{-27} \text{ kg}$$

$$= 1.145 \times 10^{-26} \text{ kg}$$

Hence,

$$\bar{\nu}_{osc} = \frac{1}{2\pi c} \times \sqrt{\frac{k}{\mu}}$$

$$\Rightarrow 2143 \text{ cm}^{-1} = \frac{1}{2\pi \times 3 \times 10^{10} \text{ cm/s}} \times \sqrt{\frac{k}{1.145 \times 10^{-26} \text{ kg}}}$$

$$\Rightarrow k = 1868 \text{ kg s}^{-2}. \quad \underline{\text{(Ans)}}$$

* How many normal modes of vibration are possible for the following molecules —

1) HBr

HBr is a linear molecule. For linear molecules,

modes of vibration $= 3N - 5$ ; $N = 2$

$= 1$

2) $O_2 \longrightarrow$ Linear

modes of vibration $= 3N - 5$ ; $N = 2$

$= 1$

3) $OCS \longrightarrow$ Linear

modes of vibration $= 3N - 5$ ; $N = 3$

$= 4$

4) $SO_2$ → Bent

modes of vibration = 3N − 6 ; N = 3

= 3

5) $BCl_3$ → Non-linear

modes of vibration = 3N − 6 ; N = 4

= 6

6) $HC \equiv CH$ → Linear

modes of vibration = 3N − 5 ; N = 4

= 7

7) $CH_4$ → Non-linear

modes of vibration = 3N − 6 ; N = 5

= 9

8) $CH_3I$ → Non-linear ; 3N − 6 = 9

9) $C_6H_6$ → non-linear

modes of vibration = 3N − 6 ; N = 12

= 30

* Explain why the C=O stretching vibration of an aldehyde gives rise to a strong absorption in IR, although the absorption due to the C=C vibration in alkene is weak. The strength of IR absorption depends on the change in dipole moment of a molecule as it vibrates. Thus the absorption due to C=O & C=C vibration are different

The C=O bond in aldehyde is a polar bond due to the significant difference in electronegativity betⁿ C & O, thus creates an observable dipole moment. Hence, when the C=O bond stretches or compresses, the dipole moment undergoes a significant change & strongly interacts with IR radiation.

On the other hand, the C=C bond in an alkene is not as polar as C=O bond since the two C atoms carry almost identical charges; exactly identical if the substituents are same on both C. Thus, C=C bond has a much smaller dipole moment or in some cases, almost no dipole moment. So it interacts weakly with IR radiation.

Therefore, the C=O stretching vibration of an aldehyde gives rise to a strong absorption in IR due to the change in dipole moment, whereas, the absorption due to C=C vibration in alkene is very weak for a negligible dipole moment change.

* How can inter & intramolecular H-bonding be identified by IR?

There are two types H-bonding – 1. Inter & 2. Intramolecular H-bonding. Both can be identified by IR spectroscopy.

The H-bonding betⁿ two different molecules is called inter-molecular H-bonding. And the H-bonding within the same molecule is called intra-molecular H-bonding.

Intermolecular H-bonding gives rise to broad bands while intra-molecular H-bonds give sharp & well defined bands.

The inter & intra-molecular H-bonds can be also identified by dilution. Intra-molecular H-bonding remains unaffected by dilution & so the absorption band also remains unaffected. Whereas, in intermolecular H-bonds, the bonds are broken on dilution & as a result there is a decrease in the bonded O-H absorption.

Thus, inter & intra-molecular H-bonds are identified by IR.

* Sketch vibrational-rotational energy levels & show the transitions.

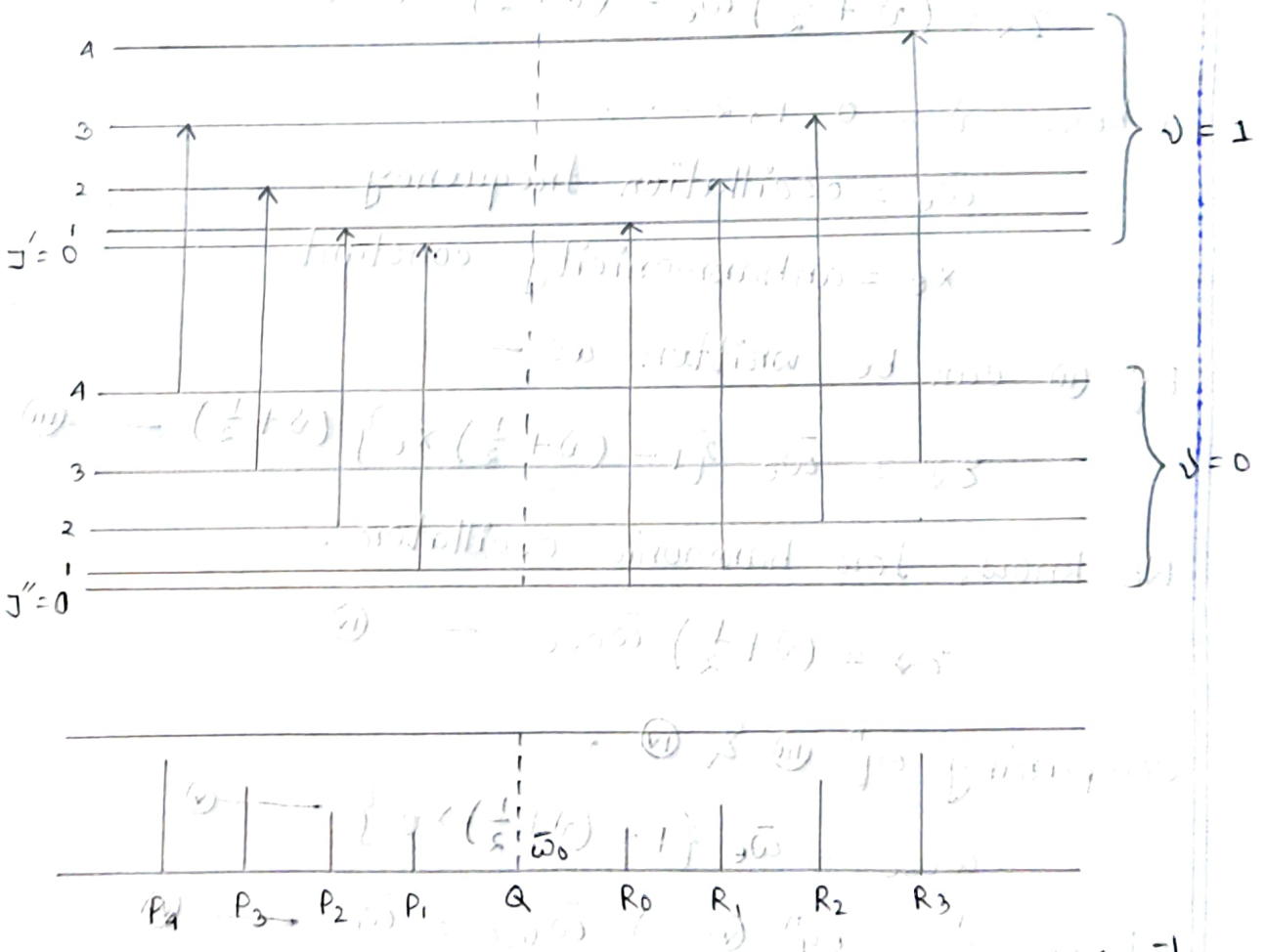



Fig: Some transitions betⁿ rotational-vibrational energy levels of a diatomic molecule, with the spectrum arising from them